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Effect of Infrared Drying on Nutrient Retention and Quality of Germinated Pea Flakes

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Abstract

This study characterized biochemical changes during pea seed (*Pisum sativum* L. cv. Aksary) germination, quantified thermolabile nutrient retention (vitamin C, thiamine, lysine) during infrared (IR) drying relative to grain temperature, optimized drying time and roller gap for maximal flake yield, and assessed physicochemical, nutritional, and microbiological quality of final flakes. Seeds were germinated (18–22 °C, 36–42 h, radicle 3–5 mm), IR dried (0.8–3.0 µm; heat flux 6.7 kW/m²; 0–180 s), and roll flaked (gap 0.3–0.8 mm) using the response surface methodology (central composite design); AOAC/ISO methods were used for nutrient analyses. Germination reduced trypsin inhibitor activity by 48% and phytic acid from 1.20 to 0.65 g/100 g dry weight, while vitamin C rose 3.9-fold and folic acid increased 47%. Nutrient losses accelerated sharply between 90 and 120 s once core temperature exceeded 70 °C. At the optimal condition (90 s; 0.6 mm roller gap), thiamine, vitamin C, and lysine retention reached 96.9%, 81.3%, and 96.8%, with 91% flake yield. This is the first study linking IR drying time, grain temperature, and nutrient retention in germinated peas, offering a science-based industrial protocol. The novelty of this study lies in defining kinetic degradation thresholds and a validated operating window for high bioactive retention with superior flake integrity.

Keywords: *Pisum sativum*; Germinated Pea; Infrared Drying; Nutrient Retention; Flake Yield; Antinutritional Factor.

1. Introduction

Pea (*Pisum sativum* L.) is one of the most widely grown grain legume crops worldwide, with annual production exceeding 13 million metric tons [1]. Legume crops provide plant-based protein, dietary fiber, and bioactive compounds, such as polyphenols and antioxidants, which are well documented for their beneficial roles in reducing the risk of many chronic non-communicable diseases at a fraction of the environmental cost of animal foods [2, 3]. Among legume crops, peas are an attractive functional ingredient source because they have a relatively well-balanced amino acid profile, a mild flavor compared to soybean or faba bean, and wide agro-climatic adaptability [4]. With the recent growing interest in plant-based diets, the RTE breakfast cereal and flake markets have emerged as drivers for incorporating legume fractions to increase protein, dietary fiber, and micronutrient density beyond traditional cereal-based formulations [5].

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One of the best bioprocessing methods to enhance the nutritional quality of grain legumes is germination. Sprouting activates endogenous phytases, endoproteases, and α -amylases, which catalyze the mobilization of storage macromolecules, release free amino acids and bioavailable minerals, and drastically reduce antinutritional factors such as phytic acid and trypsin inhibitors [6–10]. Vitamin biosynthesis is upregulated, particularly ascorbic acid, thiamine, riboflavin, pyridoxine, folate, and antioxidant compounds. This leads to the improvement of the nutritional and bioactive profiles of seeds [11–13]. All this makes germinated peas valuable raw materials for developing functional cereal-type products with an excellent amino acid score relative to the FAO/WHO reference patterns [14].

However, the nutritional advantages gained during germination can be easily lost during subsequent dehydration if inappropriate drying conditions are applied. Conventional hot-air convective drying, which typically involves moderate to high temperatures for long periods, results in substantial losses of thermolabile vitamins, protein denaturation, and Maillard-type browning that reduces lysine bioavailability [7, 15]. Such drawbacks have driven interest in alternative thermal technologies that offer rapid moisture removal with less heat damage to sensitive nutrients. Among these, infrared (IR) drying has gained attention because of its high energy transfer efficiency and selective, rapid heating of moist food matrices [16]. Unlike conventional methods, water molecules directly absorb IR radiation (0.8–3.0 μm) in the surface and subsurface layers of grains, enabling faster moisture removal and shorter processing times [17]. Recent reviews have identified IR drying as a promising compromise between energy efficiency, short process time, and quality retention compared to hot air, microwave, freezing, and vacuum drying [18, 19]. Nevertheless, if not properly controlled, the high heat flux during IR processing can cause excessive internal grain temperatures, accelerating oxidative degradation of vitamin C and Maillard condensation of lysine. Thus, IR drying faces a dual optimization challenge: under processing results in excessive moisture and poor flaking properties, while overprocessing destroys the nutrients that justify the process.

Despite these potential benefits, no study has quantitatively related IR drying time and internal grain temperature dynamics to the retention of individual thermolabile nutrients such as ascorbic acid, thiamine, and lysine in germinated peas. Consequently, there is a lack of knowledge to design dehydration protocols that balance dehydration efficiency and nutritional quality. The present study therefore aimed to fill this gap by: (I) characterizing the biochemical changes occurring during germination of *Pisum sativum* cv. Aksary regarding vitamin and amino acid profiles and anti-nutritional factors; (II) quantifying the retention of vitamin C, thiamine, and lysine as a function of IR drying time and grain surface/core temperature; (III) optimizing IR drying time and roller gap for maximum whole flake yield using response surface methodology; and (IV) evaluating the physicochemical, nutritional, and microbiological quality of the final product. The main findings were that IR drying for 90 s and a 0.6 mm roller gap secured >96% retention of thiamine and lysine, with whole flake yield reaching 91%. Furthermore, critical nutrient degradation occurred between 90 and 120 s when core temperature exceeded 70 °C, thus establishing a science-based feasible protocol for industrial production of germinated pea flakes.

2. Materials and Methods

2.1. Raw Materials

The *Pisum sativum* cv. Aksary pea seeds used in this study were bred and state-registered in Kazakhstan (harvest 2024-2025). Aksary is a yellow-seeded determinate-stem short-season variety (growing period 65-70 days) well adapted to the agroclimatic conditions of southern Kazakhstan. Table 1 summarizes the key physical and chemical characteristics of the seed lot before processing. The grain was stored under controlled conditions ($T \leq 15$ °C; relative humidity $\leq 65\%$) and was never subjected to industrial drying.

Table 1. Physical and chemical characteristics of raw pea seeds (*Pisum sativum* cv. Aksary, harvest 2024-2025)

Parameter	Value (mean $\pm$ SD)	Reference method
Bulk density (test weight), g/L	766.7 $\pm$ 1.4	ISO 7971-3:2019
Thousand-kernel weight, g	162.5 $\pm$ 0.7	ISO 520:2010
Kernel density, g/cm ³	1.31 $\pm$ 0.01	Pycnometer (toluene)
Kernel volume, mm ³	47.5 $\pm$ 0.3	Pycnometer
Moisture content, %	12.0 $\pm$ 0.1	AOAC 934.01
Crude protein (N $\times$ 6.25), g/100 g DW	24.5 $\pm$ 0.4	AOAC 920.87
Crude fat, g/100 g DW	2.1 $\pm$ 0.1	AOAC 945.16
Total starch, g/100 g DW	46.3 $\pm$ 0.8	ISO 6647-2:2011
Total dietary fiber, g/100 g DW	6.5 $\pm$ 0.3	AOAC 985.29
Ash, g/100 g DW	3.1 $\pm$ 0.1	AOAC 942.05

Note: Values are mean $\pm$ standard deviation (n = 3). DW = dry weight basis.

^oT = titratable acidity expressed as mL of 0.1 M NaOH per 100 g sample.

2.2. Germination

Before germination, seeds were washed with a neutral detergent, surface-sanitized with a sodium hypochlorite solution (200 ppm, 5 min), and rinsed three times with sterile distilled water. Seeds were then steeped in sterile distilled water (seed-to-water ratio 1:4 w/v) for 12 h at 18-22 °C. After steeping, water was drained, and seeds were

spread on filter paper in a germination chamber at 18–22 °C in the dark. Germination was allowed to proceed until the radicle reached 3–5 mm, which under these conditions required 36–42 h. Freshly germinated grain (38–42% moisture, FWB) was used immediately for the subsequent infrared drying step.

2.3. Infrared Drying

Germinated pea grains were dried in the near- and mid-infrared (IR) region (wavelength 0.8–3.0 μm) using a laboratory-scale UTZ-type dryer (Almaty Technological University, Almaty, Kazakhstan), which delivered a constant heat flux density of 6.7 kW/m² to the grain surface throughout all runs. The material ($\approx$ 0.9–1.4 kg per batch, calculated from the seed lot's bulk density, Table 1) was spread as a single layer (bed depth 1.0–1.5 cm) on a 30×40 cm tray and irradiated by a slightly larger 34×44 cm quartz-tube panel to ensure uniform heating across the tray edges. Exposure time was the only variable in this study, being tested from 0 to 180 s at 30-s intervals under this constant flux. Immediately afterward, samples were analyzed for nutrient content (Table 4) to isolate the effect of infrared exposure alone. The irradiated material was then further dried by convective hot air at 80 °C for 5–6 min to reach an equilibrium moisture content of 12–14% (total processing time 8–10 min); flakes obtained after both stages were used for all remaining quality analyses (Tables 6–10). Surface and core temperatures were monitored at each time point with a surface-contact thermocouple and a needle probe inserted into the grain's geometric center, respectively. The 90-s exposure removed about 25% of the initial water content, lowering it from 38–42% to 18–20% (wet basis); the subsequent convective step brought it down further to 12–14%.

2.4. Roll Flaking and Sieve Fractionation

After IR treatment, the partially dried germinated grain was allowed to cool to ambient temperature (22 ± 2 °C) and then flaked in a laboratory smooth-roll flaking mill fitted with cast iron rolls (diameter 400 mm, roll speed 70 rpm) [20, 21]. The roller gap was varied systematically within the range 0.3 to 0.8 mm by 0.1 mm increments, and 0.6 mm was found optimal in terms of whole-flake yield (see Section 3.3). The flaked product was mechanically separated by sieving into three fractions: (I) whole flakes, unbroken flakes with a kernel area $\geq$ two-thirds of the original and thickness equal to the set gap; (II) broken grain and grits, fragments smaller than two-thirds of the original but larger than 1 mm; and (III) flour, passing through a 1 mm aperture. Each fraction's yield was calculated as a percentage of the initial 100 g of germinated, IR-treated pea.

2.5. Determination of Antinutritional Factors

Phytate was determined colorimetrically by the method of Vaintraub & Lapteva [22]. Samples were extracted with 2.4% HCl in methanol, centrifuged, and reacted with Wade reagent (FeCl_3 in sulfosalicylic acid). The absorbance was read at 500 nm on a UV/VIS spectrophotometer (T80, PG Instruments, UK), and phytate was determined from the phytic acid standard curve.

Trypsin inhibitor activity (TIA) was determined by the method of Kakade et al. [23] using N-alpha-benzoyl-DL-arginine-p-nitroanilide (BAPA) as the chromogenic substrate. Extraction was performed in 0.01 M NaOH (pH 8.4–10.0) for 1 h for ungerminated grain and 3 h for germinated grain and flakes to ensure the full release of the inhibitor active sites. When necessary, 1% polyvinylpyrrolidone (PVP) was added to reduce the effect of phenolic compounds [24]. Absorbance was measured at 410 nm, and the results were expressed as mg of trypsin inhibited per gram of sample dry weight (mg/g DW) in accordance with the units presented in Tables 3 and 8.

Urease activity was estimated by the delta-pH method based on the AOCS Official Method Ba 9-58 [25]. The ground sample (0.20 g) was incubated with a pH 7.0 phosphate-buffered urea solution at 30 °C for 30 min. The urease index was calculated as the difference between the pH of the test solution and the substrate-free blank. Results are expressed as delta-pH units. The acceptable urease index for sufficiently processed legume products is delta-pH 0.0–0.2.

2.6. Nutritional Composition Analyses

The moisture content was gravimetrically determined after oven-drying at 105°C to a constant weight (AOAC 934.01). Crude protein was determined by the Kjeldahl method (AOAC 920.87), where digestion with concentrated H_2SO_4 was carried out in the presence of a selenium catalyst, and back-titration was performed with 0.1 M HCl. $\text{N} \times 6.25$ was used for the nitrogen-to-protein conversion factor for pea protein, as recommended by Mariotti et al. [26]. Crude fat was extracted by the Soxhlet apparatus with petroleum ether (boiling range 40–60°C) for 6 h (AOAC 945.16). Total starch was determined polarimetrically (ISO 6647-2:2011). Total dietary fiber (soluble + insoluble) was determined by the enzymatic-gravimetric method AOAC 985.29. Ash was determined gravimetrically after incineration at 550 °C for 5 h (AOAC 942.05). Titratable acidity was determined by potentiometric titration of a 1:10 (w/v) aqueous suspension with 0.1 M NaOH up to pH 8.1.

Vitamin composition (B1, B2, B6, PP, C, E, and folic acid B9) was determined by reversed-phase HPLC using a C18 column (250 × 4.6 mm, 5 μm particle size; Agilent Technologies, Santa Clara, CA, USA) with UV-DAD and fluorescence detection. The mobile phase was 0.1% trifluoroacetic acid in water (A) and acetonitrile (B) in a gradient elution mode at a flow rate of 1.0 mL/min. Vitamins were identified and quantified using external calibration curves prepared from certified reference standards (Sigma-Aldrich, Germany; purity $\geq$ 98%). Vitamin C was also confirmed

by the 2,6-dichlorophenolindophenol (DCPIP) titrimetric method (AOAC 967.21). Beta-carotene was determined spectrophotometrically at 450 nm after petroleum ether extraction and saponification.

The amino acid composition was determined by ion-exchange chromatography (IEC) with post-column ninhydrin derivatization on an automatic amino acid analyzer according to AOAC 994.12. Samples were hydrolyzed in 6 M HCl at 110 °C for 24 h under a nitrogen atmosphere; tryptophan was determined separately by alkaline hydrolysis in 4 M NaOH at 110 °C for 22 h. AAS was calculated using the FAO/WHO/UNU (2007) reference pattern for adults [27], and the limiting amino acid was considered the amino acid with the lowest AAS [28]. The mineral composition (K, Na, Ca, Mg, P, Fe, Zn, Cu) was determined by ICP-OES (AOAC 984.27) and FAAS (AOAC 975.03) after wet acid digestion in a mixture of concentrated HNO₃ and HClO₄ (3:1, v/v). Certified reference material (NIST SRM 1570a, Spinach Leaves) was added to each analytical batch for method validation, with analyte recovery between 95 and 103%.

2.7. Experimental Design for Optimization (Response Surface Methodology)

A central composite rotatable design (CCRD) was employed to optimize the combined effect of IR drying time (X_1 , coded levels -1.414, -1, 0, +1, +1.414 corresponding to 30, 60, 90, 120, 150 s) and roller gap (X_2 , coded levels -1.414, -1, 0, +1, +1.414 corresponding to 0.3, 0.4, 0.6, 0.7, 0.8 mm) on whole flake yield. The design consisted of 13 experimental runs: 4 factorial points, 4 axial points, and 5 center point replicates. The runs were randomized to minimize systematic errors. The response variable (Y , whole flake yield, %) was fitted to a second-order polynomial model. The quadratic model coefficients were estimated by multiple linear regression, and the stationary point was determined by solving the partial derivatives. The adequacy of the model was evaluated by ANOVA, coefficient of determination (R^2), and lack-of-fit test.

2.8. Statistical Analysis

Each experiment was done in triplicate ($n = 3$). Results are expressed as mean $\pm$ standard deviation (SD). Differences among treatments were tested for statistical significance using one-way ANOVA and Tukey's HSD post hoc test at $p < 0.05$. Statgraphics Centurion software (Version 19, Statgraphics Technologies Inc., The Plains, VA, USA) was used for calculations.

2.9. Flowchart of the Experimental Workflow

Figure 1 presents a flowchart summarizing the complete sequence of experimental steps performed in this study. The schematic includes preparation and germination of pea seeds (*Pisum sativum* L. cv. Aksary); infrared drying with varying treatment times; roll flaking with different roller gaps; and subsequent analyses of the final product—determination of antinutritional factors, vitamin and amino acid composition, and physicochemical and microbiological parameters.

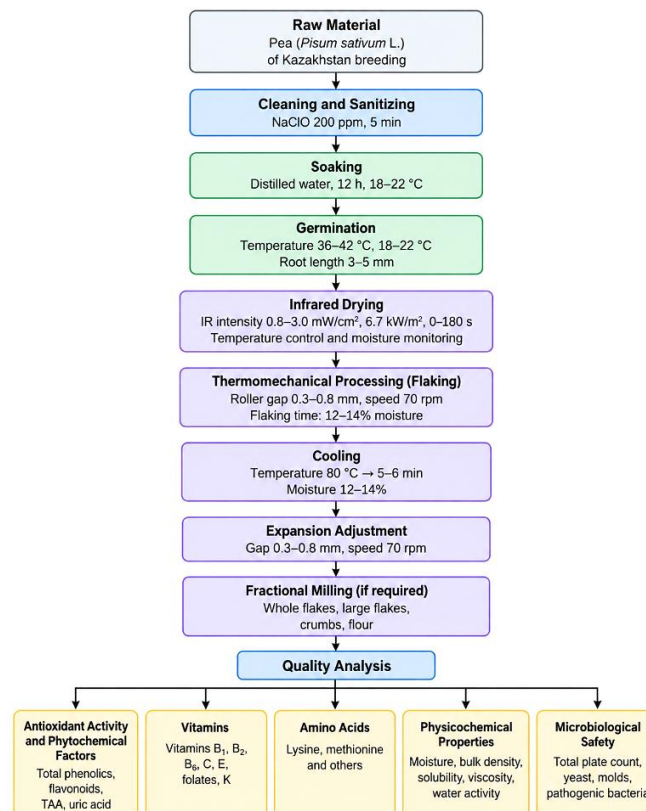


Figure 1. Flowchart of the experimental process for producing germinated pea flakes using infrared drying and roll flaking

3. Results and Discussion

3.1. Germination-Induced Biochemical Changes

Pisum sativum L. cv. Aksary germination length caused a wide range of compositional changes, increasing the grain's nutritional value. Table 2 shows the full chemical composition of the germinated grain.

Table 2. Chemical composition of germinated pea grain (*Pisum sativum* cv. Aksary, harvest 2024–2025; per 100 g fresh weight)

Parameter	Content (mean $\pm$ SD)	Change upon germination
Energy value, kcal/100 g FW	120–128	↓ 5–8 % vs. dry grain
Protein, g/100 g FW	8.5–9.2 $\pm$ 0.3	Stable; digestibility ↑
Fat, g/100 g FW	0.6–0.8 $\pm$ 0.1	Minor change
Total carbohydrates, g/100 g FW	25.0–28.0 $\pm$ 0.8	↓ via respiration
of which free sugars, g/100 g FW	4.5–6.5 $\pm$ 0.4	↑ 2–3× by starch hydrolysis
Dietary fiber, g/100 g FW	6.5–7.5 $\pm$ 0.3	Structure modified; bioavailability ↑
Vitamin C (ascorbic acid), mg/100 g FW	11.2 $\pm$ 0.7	De novo synthesis; key marker
Vitamin B ₁ (thiamine), mg/100 g FW	0.23 $\pm$ 0.02	↑ 15–25 %
Vitamin B ₂ (riboflavin), mg/100 g FW	0.12 $\pm$ 0.01	Moderate ↑
Folic acid (B ₉), µg/100 g FW	148 $\pm$ 8	↑ 1.5–2× vs. raw grain
Potassium (K), mg/100 g FW	370 $\pm$ 15	High; bioavailable form
Phosphorus (P), mg/100 g FW	165 $\pm$ 10	Released from phytate
Magnesium (Mg), mg/100 g FW	55 $\pm$ 4	Released from phytate
Calcium (Ca), mg/100 g FW	37 $\pm$ 3	Released from phytate
Iron (Fe), mg/100 g FW	2.2 $\pm$ 0.2	Bioavailability ↑
Zinc (Zn), mg/100 g FW	1.1 $\pm$ 0.1	Bioavailability ↑

Note: Values are mean $\pm$ SD (n = 3). FW = fresh weight basis. ↑ = increase; ↓ = decrease relative to raw ungerminated grain.

The most significant vitamin response was observed for ascorbic acid: in germinated grain, its content reached 11.2 $\pm$ 0.7 mg/100 g FW, i.e., 5.3 times higher than in raw seed (2.1 mg/100 g), indicating de novo biosynthesis via the L-galactose pathway during the differentiation of chloroplasts in the growing embryo axis. Vitamins of the B-group also increased: thiamine + 15.4%, riboflavin + 25.0%, pyridoxine + 18.5%, and folic acid + 46.9% compared with raw grain. The increment of folic acid is among the highest for short-cycle germination of cool-season legumes; Jribi et al. [29] reported 40–55% for germinated chickpea and lentil, which agrees well with our findings. β -Carotene was synthesized de novo (0.028 $\pm$ 0.002 mg/100 g DW; absent in control flakes) upon activation of the isoprenoid pathway in differentiating plastids [12], and tocopherol (+13.3%) and niacin (+18.8%) also increased [13].

Phytic acid decreased from 1.20 $\pm$ 0.10 to 0.65 $\pm$ 0.08 g/100 g DW (–45.8%, $p < 0.05$), and TIA from 0.50 $\pm$ 0.05 to 0.26 $\pm$ 0.03 mg/g DW (–48.0%, see Table 3).

Table 3. Changes in antinutritional factors during germination of pea grain (cv. Aksary)

Parameter	Raw grain	Germinated grain (sprout 3–5 mm)	Reduction (%)
Urease activity, Δ pH	0.06 $\pm$ 0.01 ^a	0.05 $\pm$ 0.01 ^b	16.7
Trypsin inhibitor activity (TIA), mg/g DW	0.50 $\pm$ 0.05 ^a	0.26 $\pm$ 0.03 ^b	48.0
Phytic acid, g/100 g DW	1.20 $\pm$ 0.10 ^a	0.65 $\pm$ 0.08 ^b	45.8

Note: Values are mean $\pm$ SD (n = 3). Different superscripts within the same row indicate significant differences ($p < 0.05$). TIA – trypsin inhibitor activity; DW – dry weight.

These reductions are consistent with those reported by Yılmaz Tuncel et al. [4] (TIA –44 to –50%; phytate –43 to –52%) and Wu & Shin [6] (47% TIA reduction with improved protein digestibility). The decrease in phytate is attributed to induced phytase activity, releasing mineral cations from insoluble phytate–mineral complexes [8]. The increased mineral bioavailability caused by germination agrees with Abdollahikhamene et al. [15], who found that germination of five pea varieties substantially increased Fe and Zn bioaccessibility through phytate degradation. Urease activity (Δ pH = 0.05 $\pm$ 0.01) fell within the AOCS Ba 9 58 acceptable range (0.0–0.2) [25].

3.2. Effect of Infrared Drying Duration on Nutrient Retention

The effect of IR drying time (0–180 s) on vitamin B1, vitamin C, and lysine retention, as well as the temperature on the grain surface and inside the grain, is shown in Table 4 and Figure 2. It should be noted that the data in Table 4 represent the nutrient contents after IR drying (before the subsequent convective step) and are expressed on a dry weight basis for consistency with the optimization analysis. The retention values are calculated relative to the non-dried germinated grain (time 0 = 100%).

Table 4. Effect of IR drying time on vitamin and lysine retention and grain temperature parameters

Time (s)	Vit. B ₁ (mg/100 g DW)	B ₁ ret. (%)	Vit. C (mg/100 g DW)	Vit. C ret. (%)	Lysine (g/100 g DW)	Lys. ret. (%)	Surface temp. (°C)	Internal temp. (°C)
0 (ctrl)	0.65 ± 0.01 ^a	100.0	12.8 ± 0.3 ^a	100.0	1.33 ± 0.02 ^a	100.0	22 ± 1 ^f	22 ± 1 ^f
30	0.65 ± 0.01 ^a	100.0	12.5 ± 0.3 ^a	97.7	1.33 ± 0.02 ^a	100.0	40 ± 2 ^e	22 ± 1 ^f
60	0.64 ± 0.01 ^a	98.5	11.9 ± 0.2 ^b	93.0	1.31 ± 0.02 ^a	98.9	56 ± 2 ^d	27 ± 1 ^e
90	0.63 ± 0.01 ^b	96.9	10.4 ± 0.2 ^c	81.3	1.29 ± 0.01 ^b	96.8	75 ± 3 ^c	40 ± 2 ^d
120	0.60 ± 0.02 ^c	92.3	8.2 ± 0.2 ^d	64.1	1.17 ± 0.02 ^c	88.2	96 ± 3 ^b	73 ± 2 ^c
150	0.56 ± 0.02 ^d	86.2	5.1 ± 0.1 ^e	39.8	1.01 ± 0.02 ^d	76.3	112 ± 4 ^a	94 ± 3 ^b
180	0.51 ± 0.02 ^e	78.5	2.3 ± 0.1 ^f	18.0	0.84 ± 0.01 ^e	63.4	125 ± 5 ^a	108 ± 4 ^a

Note: Values are expressed on DW basis; mean ± SD (n = 3). Different superscripts within the same column indicate significant difference (p < 0.05).

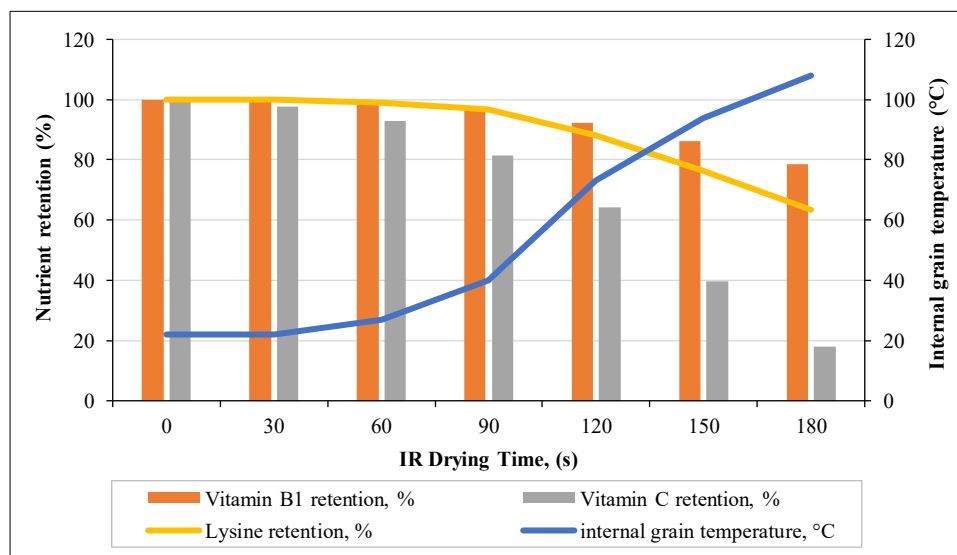


Figure 2. Retention of vitamin B₁, vitamin C and lysine and internal grain temperature as a function of IR drying time in germinated pea (*Pisum sativum* L. cv. Aksary)

With increasing IR exposure, the grain surface and core temperatures increased non-linearly. After 30 s, the surface temperature was 40 ± 2°C, and the grain center was 22 ± 1°C, indicating poor thermal conductivity of the pea kernel and the localized energy absorption at the grain surface, typical of early IR heating [16, 30]. At 90 s, the surface temperature was 75 ± 3 °C, and the core was 40 ± 2 °C; after 180 s, the surface and core temperatures were 125 ± 5 °C and 108 ± 4 °C, respectively. The critical degradation time was identified as 90–120 s when the grain core exceeded 70°C. Chatchavanthatri et al. [31] reported a similar temperature threshold for germinated brown rice, where nutrient loss accelerated when core temperature exceeded 70–75°C. The surface core gradients of 30–40°C within the first 1–2 min of IR exposure are also in line with reports for wheat and paddy [16, 30].

Thiamine retention was >96% for all sampling times up to 90 s. Significant losses first occurred at 120 s (92.3%; p < 0.05, Tukey HSD), further declining to 78.5% at 180 s. In near-neutral food systems, thiamine thermal degradation occurs mainly by heterolytic fission of the thiazolium ring, with the reaction rate increasing steeply above 80–90 °C; the core temperature at 90 s (40 °C) was thus well below the thermal threshold. Marchini et al. [32] observed similar behavior for sorghum, with thiamine losses below 6% at core temperatures below 70°C and accelerated degradation above this temperature. Sarjerao et al. [33] reported 74–78% thiamine retention after 60 min of hot air drying at 70 °C for germinated mung bean, highlighting the advantage of the short IR regime.

Vitamin C was the most heat-sensitive compound, showing a statistically significant loss from 60 s onwards (93.0%; p < 0.05). After 90 s, 81.3% remained; this decreased to 64.1% after 120 s, and only 18.0% remained after 180 s. Ascorbic acid is degraded by two competing routes: aerobic oxidation to 2,3-diketogulonate at elevated temperatures and acid-catalyzed anaerobic dehydration to furfural derivatives. Both routes are accelerated during IR

treatment since oxidation kinetics increase as temperature increases, and oxygen diffusivity in the grain surface layer increases with progressive moisture loss [34]. The 81.3% retention at 90 s, with a surface temperature of 75 °C, compares favorably with the 44–62% retention for sprouted chickpea during conventional drying at 80 °C for 60 min [4] and fits well with the IR kinetic data reviewed by Huang et al. The degradation pattern followed first-order kinetics, as demonstrated by Giannakourou & Taoukis [35] for ascorbic acid in plant-based matrices, with rate constants strongly dependent on temperature and water activity. Goharpour et al. [36] also reported that IR drying was superior to hot air and microwave drying for preserving bioactive compounds in ground sprouted chickpeas, with significantly higher phenolic content and antioxidant capacity, supporting the role of IR technology in processing sprouted legumes.

Lysine retention followed a different pattern from that of the vitamins: losses were insignificant up to 90 s (<4%; $p > 0.05$ vs. control), but then dropped sharply to 88.2% at 120 s and 63.4% at 180 s. The threshold-driven kinetics are typical of Maillard condensation, in which reactive intermediates accumulate until temperature and water activity conditions allow enhanced propagation of the reaction [37, 38]. Germination provides the carbonyl substrate through α -amylase activity, which releases free glucose and maltose from starch reserves [13]. When the grain core exceeds 70°C and moisture decreases, water activity approaches the range of 0.6–0.8 at which the Maillard rate is maximal [37]. The resulting Amadori products irreversibly block the ϵ -amino groups of lysine [29]. Marchini et al. [32] reported an identical temperature threshold for lysine in post-sprouting dried sorghum, with losses occurring above 70–75 °C at the core. Methionine also decreased in germinated flakes (–22.6% vs. raw grain) because of its involvement in the Maillard reaction and thermal aerobic oxidation to methionine sulfoxide [37]. Najib et al. [39] showed that microwave-assisted IR drying of lentils reduces drying time by 96% compared to conventional drying while maintaining quality, suggesting that hybrid systems could be explored beyond the single-pass IR studied herein.

Lysine loss was only 3.2% at the selected optimum of 90 s at a core temperature of 40°C. With this condition, more than 80% of vitamin C, more than 97% of thiamine, and more than 97% of lysine were retained, with grain moisture lowered to 18–20%, making this the optimal single-stage IR treatment. These results concur with Su et al. [40], who found that germination followed by short-duration IR drying of naked barley resulted in significantly better retention of starch functionality and protein quality than conventional hot air drying, indicating that short IR pretreatment is the optimal dehydration approach for germinated grain legumes and cereals.

3.3. Effect of Roller Gap on Whole-Flake Yield

Table 5 and Figure 3 summarize the distribution of whole flakes, broken grain, and flour over the roller gap (at fixed IR time 90 s). The gap's effect on the whole-flake yield was highly significant (one-way ANOVA, $p < 0.001$), but it was not linear and reached a maximum at 0.6 mm (91 ± 1.5% whole flakes; total by-products 9%).

Table 5. Effect of the roller gap on the product fractions (IR drying time fixed at 90 s; % of 100 g batch)

Gap (mm)	Whole flakes (%)	Broken grain & grits (%)	Flour (<1 mm) (%)	Total by-products (%)
0.3	55 ± 1.0 ^c	38 ± 1.5 ^a	7 ± 1.0 ^a	45
0.4	78 ± 1.0 ^d	20 ± 1.0 ^b	2 ± 0.5 ^b	22
0.5	83 ± 1.0 ^c	14 ± 1.0 ^c	3 ± 0.5 ^b	17
0.6	91 ± 1.5 ^a	7 ± 1.0 ^d	2 ± 0.5 ^b	9
0.7	82 ± 0.5 ^c	15 ± 1.0 ^c	3 ± 0.5 ^b	18
0.8	75 ± 0.5 ^d	22 ± 1.5 ^b	3 ± 0.3 ^b	25

Note: Values are means ± SD (n = 3). Different superscripts within the same column indicate significant differences ($p < 0.05$).

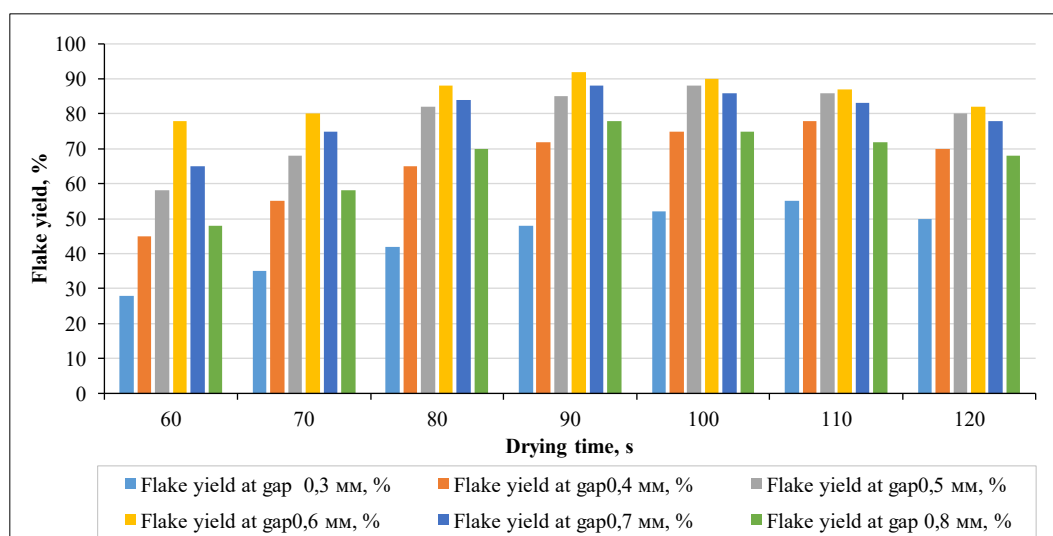


Figure 3. Effect of IR drying time and roller gap on whole-flake yield of germinated pea (*Pisum sativum* L. cv. Aksary)

At gaps below 0.5 mm, excessive compressive and shear stresses cause grain fragmentation and adhesion to the rolls, raising the flour fraction to 7% at 0.3 mm and total by-products to 45%. At gaps over 0.7 mm, the kernels are insufficiently deformed and, therefore, only partly flattened to a thick-flake grade, reducing the whole-flake yield to 82-75%. Pre-IR treatment for less than 80 s leaves grain moisture above 22%, causing adhesion to the rolls, whereas more than 100 s of pre-treatment makes the grains brittle. The moisture range of 18-20%, obtained after 90 s of IR treatment, fits within the optimal moisture range of 14-22%, as defined by Cork et al. [20] for desi-type pulses, and the narrow process window between inadequate and excessive moisture removal is well confirmed. Lignicka & Galoburda [41] also showed that an optimal combination of pre drying and gap settings is necessary to achieve the target flake hardness, brittleness, and porosity simultaneously.

The interaction between IR drying time and roller gap, and its effect on whole-flake yield, was further investigated using response surface methodology. Response Surface Methodology was used to study the combined effect of IR drying time and roller gap on whole-flake yield based on a central composite rotatable design. The fitted second-order polynomial regression model for germinated peas is:

$$Y = 91.0 - 1.014X_1 - 1.628X_2 - 14.512X_1^2 + 5.000X_1X_2 - 8.862X_2^2 \quad (1)$$

where, Y is the predicted whole-flake yield (%), X_1 is the coded IR drying time, and X_2 is the coded roller gap.

The model was highly significant ($p < 0.001$) with $R^2 = 0.97$ and adjusted $R^2 = 0.94$, indicating that 97% of the variability in flake yield was explained by the model. The variance inflation factor (VIF) did not exceed 1.225 for any term, confirming no relevant multicollinearity. The quadratic terms X_1^2 and X_2^2 had the highest absolute effect -14.51 and -8.86 , respectively, indicating the existence of a well-pronounced maximum within the domain of the experiment. Since both quadratic coefficients have negative signs, the response surface is dome-shaped, with the yield decreasing in all directions away from the stationary point. The stationary point, a maximum as proven by the negative eigenvalues of the Hessian matrix, was found at $\tau = 90$ s and $a = 0.60$ mm, giving a predicted whole-flake yield of 91.0%, which is in very good agreement with the experimentally observed value of $91.0 \pm 1.5\%$ obtained from replicated center-point runs ($n = 5$). Analysis of the steepest ascent path showed that the optimum is robust; however, increasing IR drying time from 90 to 95 s at proportionally reduced roller gap dropped the predicted yield from 91.0% to 82.8%, confirming a very narrow process window and the sensitivity of flake yield to deviations from optimum conditions (Figures 4 and 5).

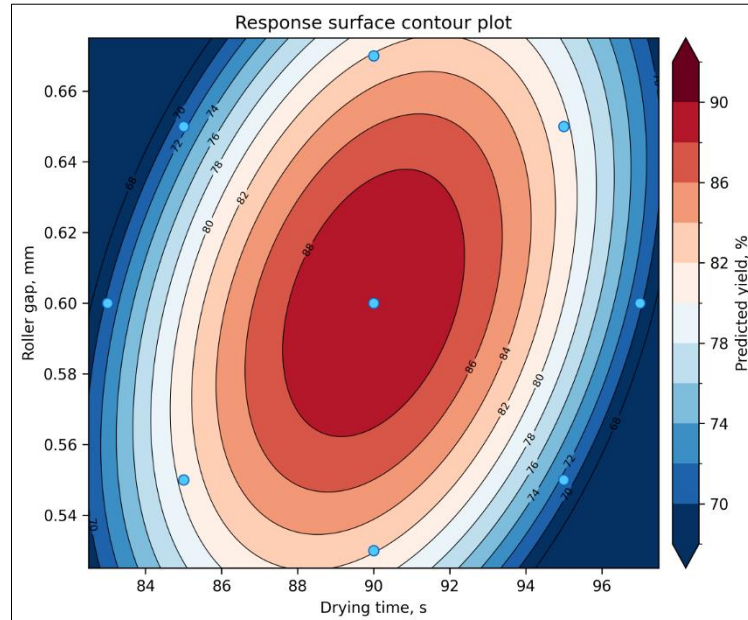


Figure 4. Response surface contour plot of the predicted whole-flake yield (%) as a function of IR drying time (s) and roller gap (mm) for germinated pea (*Pisum sativum* L. cv. Aksary)

Contour lines show iso-yield lines at 2% intervals; the color scale from deep blue = 70% to dark red = 90% shows the magnitude of the predicted response over the experimental domain. Blue dots represent the experimental run points of the central composite design. The stationary point of the fitted quadratic model is found at a drying time of 90 s and a roller gap of 0.60 mm (predicted yield 91%) and is located at the center of the innermost dark-red ellipse. Moving away from this optimum in any direction of the factors results in a steady decline in whole-flake yield, as indicated by the increasingly blue contours. The asymmetry of the ellipses along the two axes reflects the greater sensitivity of the yield to changes in roller gap than to changes in drying time within the investigated range.

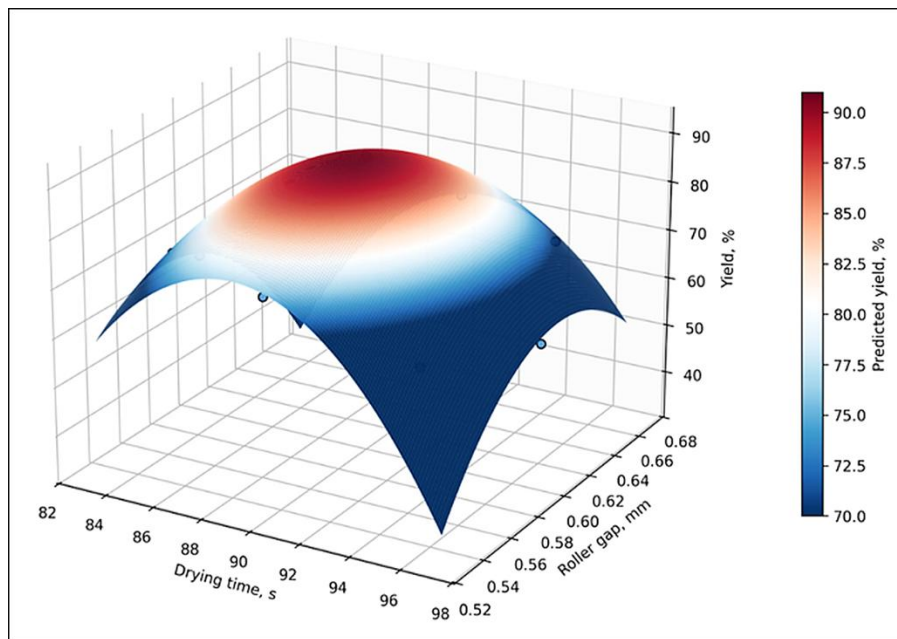


Figure 5. Estimated three-dimensional response surface for predicted whole-flake yield (%) as a function of IR drying time (s) and roller gap (mm) for germinated pea (*Pisum sativum* L. cv. Aksary)

The surface was fitted with a full second-order polynomial response surface model; the color scale from dark blue = 70% to dark red $\geq 90\%$ represents the predicted yield. Blue dots are the experimental values observed and projected on the surface. There is one global well-defined maximum, located at ~ 90 s drying time and 0.60 mm roller gap, where the surface has its summit (predicted yield 91%). The surface descends steeply towards the four corners of the factor space, so no saddle-point behavior is observed, and the high-yield region is limited and well defined. This topology is smooth and dome-shaped, evidencing the strong interaction between the two process variables, with optimal yield only achieved when drying time and roller gap were set simultaneously close to their optimum levels.

3.4. Nutritional Quality of Germinated Pea Flakes

The vitamin composition of raw pea grain, germinated pea flakes obtained at the selected process conditions (IR drying 90 s, roller gap 0.6 mm), and control flakes obtained without germination is presented in Table 6.

Table 6. Vitamin composition of pea samples (mg/100 g DW)

Vitamin	Pea grain	Germinated pea flakes	Control flakes
B ₁ (thiamine)	0.52 $\pm$ 0.02 ^b	0.60 $\pm$ 0.03 ^a	0.35 $\pm$ 0.02 ^c
B ₂ (riboflavin)	0.12 $\pm$ 0.01 ^b	0.15 $\pm$ 0.01 ^a	0.06 $\pm$ 0.01 ^c
B ₆ (pyridoxine)	0.27 $\pm$ 0.01 ^b	0.32 $\pm$ 0.02 ^a	0.11 $\pm$ 0.01 ^c
B ₉ (folic acid), $\mu\text{g}/100$ g DW	16.0 $\pm$ 0.5 ^b	23.5 $\pm$ 0.7 ^a	8.5 $\pm$ 0.4 ^c
Vitamin C	2.1 $\pm$ 0.1 ^b	8.2 $\pm$ 0.3 ^a	0.3 $\pm$ 0.1 ^c
PP (niacin)	2.4 $\pm$ 0.1 ^b	2.85 $\pm$ 0.1 ^a	1.35 $\pm$ 0.1 ^c
Vitamin E (tocopherol)	7.5 $\pm$ 0.2 ^b	8.50 $\pm$ 0.3 ^a	5.9 $\pm$ 0.2 ^c
β -carotene	0.015 $\pm$ 0.001 ^b	0.028 $\pm$ 0.002 ^a	ND ^c

Note: Values are mean $\pm$ SD ($n = 3$; folic acid in $\mu\text{g}/100$ g DW; others in mg/100 g DW). Different superscripts within the same row indicate significant differences ($p < 0.05$). ND – not detected.

Vitamin C in germinated flakes (8.2 $\pm$ 0.3 mg/100 g DW) was 3.9 times higher than in the raw grain and 27 times higher than in the control flakes. The residual 81% retention against the germinated grain baseline reflected the IR drying step (Section 3.2). Increases over raw grain for the B-group vitamins were as follows: thiamine +15.4%, riboflavin +25.0%, pyridoxine +18.5%, and folic acid +46.9%, all significantly higher than in control flakes ($p < 0.05$). The folic acid increase fell within the 40–55% range reported for germinated chickpea and lentil [29]. High retention of thiamine (96.9% at 90 s) contrasts with reported losses in high moisture extrusion of pea ingredients [42], thus positioning short-duration IR treatment as a gentler thermal alternative for retaining B group vitamins in legume-based products. Table 7 shows the amino acid composition of the three pea samples.

Table 7. Amino acid composition of pea samples (mg/100 g DW)

Amino acid	Pea grain	Germinated pea flakes	Control flakes
Valine	1520.5 ^b	1610.0 ^a	1400.2 ^c
Leucine	2310.2 ^b	2425.8 ^a	2030.5 ^c
Isoleucine	1530.8 ^b	1630.5 ^a	1040.1 ^c
Lysine	1480.3 ^a	1330.0 ^b	1250.4 ^c
Methionine	200.5 ^a	155.2 ^b	112.3 ^c
Threonine	950.7 ^b	1105.5 ^a	990.8 ^b
Phenylalanine	1580.4 ^b	1900.7 ^a	1701.9 ^c
Σ Essential amino acids	9573.4 ^b	10157.7 ^a	8526.2 ^c
Aspartic acid	2680.1 ^b	2850.4 ^a	2601.3 ^c
Glutamic acid	3750.8 ^a	3400.5 ^c	3480.6 ^b
Serine	1150.2 ^b	1395.8 ^a	990.5 ^c
Glycine	1210.5 ^b	1590.3 ^a	1070.1 ^c
Alanine	1170.3 ^a	1010.2 ^c	1065.4 ^b
Arginine	2005.6 ^a	1960.8 ^b	1870.9 ^c
Proline	780.4 ^b	825.1 ^a	745.8 ^c
Cysteine	295.0 ^b	310.5 ^a	195.2 ^c
Histidine	750.1 ^b	1175.0 ^a	730.1 ^c
Tyrosine	870.5 ^a	855.0 ^b	895.0 ^a
Σ Non-essential amino acids	14663.5 ^b	15373.6 ^a	13644.9 ^c
Total amino acids	24236.9^b	25531.3^a	22171.1^c

Note: Values are means of three replicates (n = 3). Different superscripts within the same row indicate significant differences (p < 0.05).

Total amino acid content in the germinated flakes (25,531 mg/100 g DW) was 5.3% higher than in the raw grain (24,237) and 15.2% higher than in control flakes (22,171), while essential amino acids increased by 6.1% and 19.1%, respectively. Such increases would be expected due to the proteolytic mobilization of storage globulins (legumin, vicilin) by endopeptidase and carboxypeptidase activities that occur during germination [43]. Lysine (−10.2 %) and methionine (−22.6 %) were exceptions, as noted in Section 3.2. The amino acid score (AAS) against the FAO/WHO/UNU (2007) adult reference pattern [27] showed methionine + cysteine (Met+Cys) to be the first limiting pair in all samples—a consequence of the typical sulfur amino acid deficit in the faba bean, which Guillin et al. [14] demonstrated in commercial pea protein isolates with AAS 81–96%. Met+Cys AAS: raw grain 92%, germinated flakes 88%, control flakes 57%. All other essential amino acids scored > 100% of the reference in both flake types. The 54% relative gain in AAS between control (57%) and germinated (88%) flakes quantifies a clear nutritional gain from the germination step. From an industrial point of view, the optimum IR drying regime established (90 s, 0.8–3.0 μm, 6.7 kW/m²) corresponds to 8–10 min total drying time, including the final convective step, against 60–120 min for hot-air drying to equivalent moisture endpoints. The resultant germinated pea flakes contain 24.0 g of protein and 6.5 g of dietary fiber per 100 g DW, cook in 3 min, and show a sulfur amino acid score of 88%, substantially above commercial cereal-based breakfast flakes and, therefore, nutritionally differentiated functional food ingredients.

3.5. Product Quality and Safety

Tables 8 to 10 present the antinutritional factors of raw grain and germinated pea flakes, energy and proximate composition, and physicochemical and microbiological characteristics.

Table 8. Antinutritional factors in raw pea grain and germinated pea flakes

Parameter	Raw pea grain	Germinated pea flakes
Urease activity, ΔpH	0.06 ± 0.01 ^a	0.05 ± 0.01 ^b
Trypsin inhibitor activity, mg/g DW	0.50 ± 0.05 ^a	0.26 ± 0.03 ^b

Note: Data are presented as mean ± SD (n = 3). Different superscripts within the same row indicate significant differences (p < 0.05).

Table 9. Energy and proximate composition of pea flake products

Sample	Energy (kcal/100 g)	Protein (g/100 g)	Lipids (g/100 g)	Digestible carbohydrates (g/100 g)	Dietary fiber (g/100 g)
Germinated pea flakes	295 ± 5 ^b	24.0 ± 0.5 ^a	2.1 ± 0.1 ^a	50.5 ± 1.2 ^b	6.5 ± 0.4 ^a
Control pea flakes	310 ± 5 ^a	24.4 ± 0.6 ^a	1.8 ± 0.1 ^b	59.5 ± 1.5 ^a	1.4 ± 0.2 ^b

Note: Values are mean ± SD (n = 3; DW basis). Different superscripts within the same column indicate significant differences (p < 0.05). Energy calculated using Atwater coefficients (4 kcal/g protein; 4 kcal/g digestible carbohydrate; 9 kcal/g lipid; 2 kcal/g dietary fiber).

Table 10. Physicochemical and microbiological characteristics of germinated pea flakes

Parameter	Result
<i>Sensory characteristics</i>	
Appearance and colour	Oval flakes; light cream to golden-yellow
Odour and taste	Clean, characteristic, without off-flavors
<i>Physicochemical properties</i>	
Moisture content, %	11.5 ± 0.7
Titrateable acidity, °T	2.8 ± 0.2
Cooking time, min	3.0 ± 0.5
Foreign impurities, %	Not detected
Metal-magnetic particles (≤0.3 mm), mg/kg	<0.5
<i>Microbiological parameters</i>	
Total viable count, CFU/g	(1.8 ± 0.4) × 10 ²
Molds, CFU/g	<10 (not detected)
Yeasts, CFU/g	<10 (not detected)
Coliforms in 0.01 g	Not detected
Salmonella spp. in 25 g	Not detected
Bacillus cereus in 0.1 g	Not detected

Note: Data are presented as mean ± SD (n = 3).

Urease activity decreased by 16.7%, and TIA by 48% in germinated flakes relative to raw grain; the urease index ($\Delta\text{pH} = 0.05$) falls within the AOCS Ba 9 58 range (0.0–0.2) [25], thus validating the product's safety. Based on dose response data reviewed by Samtiya et al. [5], the 48% TIA reduction is expected to significantly increase protein digestibility in vivo. Rizvi et al. [44] reported a 28.34% increase in in vitro protein digestibility of pigeon pea following germination, providing a directly comparable benchmark for the improvement in protein nutritional quality.

The germinated flakes contained 24.0 ± 0.5 g protein and 6.5 ± 0.4 g dietary fiber per 100 g DW. The protein was not significantly different from that of the control flakes (24.4 g, p > 0.05), consistent with the DW-concentrating effect of respiratory carbohydrate loss during sprouting [2]. The dietary fiber content in germinated flakes (6.5 g) was more than four times higher than that in the control flakes (1.4 g, p < 0.05). This increase is consistent with previous findings showing that germination can significantly modify and increase dietary fiber fractions in peas [45]. Moisture (11.5%) was within the <14% range, which corresponded to a water activity below the threshold for mold growth. Titrateable acidity (2.8 °T) showed no signs of rancidity. The 3-min cooking time makes the product instant/ready-to-cook, consistent with the optimal pre-conditioning of pulse flakes. Total viable count (1.8 × 10² CFU/g) matched values <5 × 10² CFU/g reported by Galal et al. [46] for cereal flakes subjected to thermal processing at >70 °C. Molds, yeasts, coliforms, Salmonella spp., and B. cereus were absent, confirming that hypochlorite sanitization during germination, combined with IR-convective drying, ensures microbiological safety without the need for additional preservation measures.

Overall, 24 g protein, 6.5 g dietary fiber per 100 g DW, AAS 88% for the limiting sulfur amino acid pair, 3.9-fold vitamin C enrichment, 3-min cooking time, and complete microbiological safety position germinated pea flakes as a functional breakfast product nutritionally differentiated from typical cereal-based breakfast foods and able to fill a documented gap [47]. This is also reflected in the growing number of studies on the demand for germinated pea flakes as a functional breakfast ingredient. The global plant-based protein market size will cross \$27 billion by 2030, and germinated legumes are one of the key sources of value-added ingredients that combine high protein quality with a low level of antinutritional factors.

4. Conclusion

This study established for the first time a quantitative relationship between IR drying duration, internal grain temperature dynamics, and thermolabile nutrient retention in germinated pea (*Pisum sativum* L. cv. Aksary), and translated these relationships into an integrated germination–IR drying–flaking protocol directly relevant to industry. By linking real-time thermal kinetics to nutrient degradation pathways, the work moves beyond empirical drying recommendations toward a mechanistically grounded processing framework.

The main findings are summarized as follows: (1) Germination (36–42 h, 18–22°C) reduced phytic acid by 45.8% and trypsin inhibitor activity by 48.0%, while increasing vitamin C 3.9-fold and folic acid by 47%, substantially improving the nutritional profile of the raw grain before thermal processing. (2) IR drying kinetic data revealed a critical threshold at 90–120 s, at which point grain core temperature exceeded 70 °C; beyond this threshold, Maillard-mediated lysine loss and ascorbic acid degradation accelerated sharply, indicating that drying time control, rather than average exposure conditions, is the decisive process variable. (3) Using response surface methodology, the optimal IR drying time (90 s) and roller gap (0.6 mm) were identified, yielding a whole flake fraction of 91% together with retention of thiamine (96.9%), vitamin C (81.3%), and lysine (96.8%), confirming that short, high-flux IR exposure preserves thermolabile nutrients far more effectively than prolonged conventional drying. (4) The final germinated pea flakes contained 24.0 g protein and 6.5 g dietary fiber per 100 g DW, required only a 3-min cooking time, and met all microbiological safety criteria, supporting their suitability as a ready-to-cook functional food ingredient. The amino acid score for sulfur-containing amino acids improved from 57% (control flakes) to 88% (germinated flakes), demonstrating a clear nutritional advantage over conventional cereal-based products and highlighting germination as a low-cost strategy for correcting the limiting amino acid profile typical of legumes.

Collectively, these results demonstrate that short-duration IR pre-drying is a superior dehydration method for preserving the nutritional gains of germination, compared with conventional hot-air drying, and provides a science-based, industrially scalable process for producing high-quality germinated pea flakes, giving manufacturers a validated basis for scale-up while safeguarding nutrient density.

4.1. Limitations and Future Study

The present study was conducted under strictly controlled laboratory conditions using a single pea cultivar (cv. Aksary, harvest 2024–2025), which limits the extrapolation of the established IR drying kinetics and nutrient retention parameters to other cultivars or growing seasons. Flaking optimization was performed at laboratory scale using a smooth-roll mill (roll diameter 400 mm); scale-up to pilot or industrial equipment may necessitate re-optimization of the roller gap and IR flux density. The microbiological safety assessment covered freshly produced flakes only; product stability during storage, water activity dynamics, and lipid oxidation kinetics were not investigated. Sensory profiling (texture, flavor, and aroma), as well as in vivo or in vitro assessment of the improved amino acid score and mineral fraction bioavailability, were outside the scope of this work. Future research should address: (i) validation of the germination–IR drying protocol across multiple cultivars and growing seasons; (ii) pilot-scale technical feasibility and energy efficiency assessment; (iii) storage stability under both controlled and ambient conditions; (iv) consumer sensory testing; and (v) in vitro protein digestibility and mineral bioavailability of the finished germinated pea flakes relative to conventional cereal-based products.

5. Declarations

5.1. Author Contributions

Conceptualization, M.Z.; methodology, M.Z., N.O., and N.P.; investigation, M.Z., N.O., and M.B.; formal analysis, M.Z. and N.P.; data curation, M.Z., N.O., and M.B.; writing—original draft preparation, M.Z.; writing—review and editing, N.O., N.P., B.K., and G.Y.; visualization, M.Z. and M.B.; validation, N.O. and N.P.; supervision, B.K. and G.Y.; project administration, B.K. and G.Y.; funding acquisition, M.Z. All authors have read and agreed to the published version of the manuscript.

5.2. Data Availability Statement

The data presented in this study are available in the article.

5.3. Funding

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5.5. Declaration of Competing Interest

The authors declare that there are no conflicts of interest concerning the publication of this manuscript. Furthermore, all ethical considerations, including plagiarism, informed consent, misconduct, data fabrication and/or falsification, double publication and/or submission, and redundancies have been completely observed by the authors.

6. References

- [1] FAO. (2025). Pea Production — UN FAO: Production: Crops and Livestock Products. Food and Agriculture Organization of the United Nations, Rome, Italy. Available online: <https://archive.ourworldindata.org/20250718-111234/grapher/pea-production.html> (accessed on August 2026).
- [2] Hall, C., Hillen, C., & Garden Robinson, J. (2017). Composition, nutritional value, and health benefits of pulses. *Cereal Chemistry*, 94(1), 11–31. doi:10.1094/CCHEM-06-16-0163-FI.
- [3] Ramírez-Ojeda, A. M., Moreno-Rojas, R., & Cámara-Martos, F. (2018). Mineral and trace element content in legumes (lentils, chickpeas and beans): Bioaccessibility and probabilistic assessment of dietary intake. *Journal of Food Composition and Analysis*, 73, 17–28. doi:10.1016/j.jfca.2018.01.010.
- [4] Yılmaz Tuncel, N., Polat Kaya, H., Sakarya, F. B., Andaç, A. E., Korkmaz, F., & Ozkan, G. (2025). The effect of germination on antinutritional components, in vitro starch and protein digestibility, content, and bioaccessibility of phenolics and antioxidants of some pulses. *Food Science & Nutrition*, e70103. doi:10.1002/fsn3.70103.
- [5] Samtiya, M., Aluko, R. E., & Dhewa, T. (2020). Plant food antinutritional factors and their reduction strategies: An overview. *Food Production, Processing and Nutrition*, 2, 6. doi:10.1186/s43014-020-0020-5.
- [6] Wu, Y., & Shin, W.-S. (2025). Germination-induced changes in the nutritional, bioactive, and digestive properties of Lima bean (*Phaseolus lunatus* L.). *Foods*, 14, 2123. doi:10.3390/foods14122123.
- [7] Peñaranda, J. D., Bueno, M., Álvarez, F., Pérez, P. D., & Pérezabad, L. (2021). Sprouted grains in product development: Case studies on sprouted wheat baking flours and fermented beverages. *International Journal of Gastronomy and Food Science*, 25, 100375. doi:10.1016/j.ijgfs.2021.100375.
- [8] Zaw, T. O. K., Anbananthan, K. S. M., Muthaiyah, S., Balasubramaniam, B., Mohammad, S., Yusoff, Y., & Kalid, K. S. (2025). Adopting TOGAF Framework for Sustainable and Scalable Robusta Coffee Leaf Rust Management. *Emerging Science Journal*, 9(3), 1308–1321. doi:10.28991/ESJ-2025-09-03-010.
- [9] Oghbaei, M., & Prakash, J. (2016). Effect of primary processing of cereals and legumes on nutritional quality: A comprehensive review. *Cogent Food & Agriculture*, 2(1), 1136015. doi:10.1080/23311932.2015.1136015.
- [10] Akkad, R., Buchko, A., Johnston, S. P., Han, J., House, J. D., & Curtis, J. M. (2021). Sprouting improves the flavour quality of faba bean flours. *Food Chemistry*, 364, 130355. doi:10.1016/j.foodchem.2021.130355.
- [11] Bartkiene, E., Lele, V., & Ruzauskas, M. (2022). Sprouting and fermentation of legumes as effective strategies to improve their nutritional and bioactive properties. *Foods*, 11(19), 3047. doi:10.3390/foods11193047.
- [12] Grela, E. R., Kowalczyk-Vasilev, E., Klebaniuk, R., & Woźniakowska, A. (2022). Carotenoid and tocopherol content in germinated legume seeds. *Antioxidants*, 11(4), 644. doi:10.3390/antiox11040644.
- [13] Lemmens, E., Moroni, A. V., Pagand, J., Heirbaut, P., Ritala, A., & Karlen, Y. (2019). Impact of cereal seed sprouting on its nutritional and technological properties: A critical review. *Comprehensive Reviews in Food Science and Food Safety*, 18(1), 305–328. doi:10.1111/1541-4337.12414.
- [14] Guillin, F. M., Gaudichon, C., & Guérin-Deremaux, L. (2022). Dietary protein quality and amino acid scoring of pea protein. *Frontiers in Nutrition*, 9, 906565. doi:10.3389/fnut.2022.906565.
- [15] Abdollahikhamene, H., Kazemzadeh Pournaki, S., & Hall, C. (2025). Effects of germination on the nutritional profile of five distinct pea varieties. *Molecules*, 30, 3114. doi:10.3390/molecules30153114.
- [16] Huang, D., Yang, P., Tang, X., Luo, L., & Sundén, B. (2021). Application of infrared radiation in the drying of food products. *Trends in Food Science & Technology*, 110, 765–777. doi:10.1016/j.tifs.2021.02.039.
- [17] Aboud, S. A., Altemimi, A. B., Al-Hilphy, R. S., Yi-Chen, L., & Cacciola, F. (2019). A comprehensive review on infrared heating applications in food processing. *Molecules*, 24, 4125. doi:10.3390/molecules24224125.

- [18] Salehi, F. (2020). Recent applications and potential of infrared dryer systems for drying various agricultural products: A review. *International Journal of Fruit Science*, 20, 586–602. doi:10.1080/15538362.2020.1717566.
- [19] Yu, P., Zhu, W., Shen, C., Qiao, Y., Zhang, W., & Zhu, Y. (2025). Current status of grain drying technology and equipment development: A review. *Foods*, 14, 2426. doi:10.3390/foods14142426.
- [20] Cork, S. D., Blanchard, C. L., Mawson, A. J., & Farahnaky, A. (2025). Desi chickpea flaking: Impact of processing parameters on flake quality. *Journal of Food Engineering*, 386, 112256. doi:10.1016/j.jfoodeng.2024.112256.
- [21] Cork, S. D., Agboola, S. O., Kasapis, S., & Gidley, M. J. (2022). Pulse flaking: Opportunities and challenges, a review. *Comprehensive Reviews in Food Science and Food Safety*, 21(3), 2873–2897. doi:10.1111/1541-4337.12931.
- [22] Vaintraub, I. A., & Lapteva, N. A. (1988). Colorimetric determination of phytate in unpurified extracts of seed and the products of their processing. *Analytical Biochemistry*, 175(1), 227–230. doi:10.1016/0003-2697(88)90382-X.
- [23] Kakade, M. L., Rackis, J. J., McGhee, J. E., & Puski, G. (1974). Determination of trypsin inhibitor activity of soy products: A collaborative analysis of an improved procedure. *Cereal Chemistry*, 51(3), 376–382.
- [24] Valdebouze, P., Bergeron, E., Gaborit, T., & Delort-Laval, J. (1980). Content and distribution of trypsin inhibitors and hemagglutinins in some legume seeds. *Canadian Journal of Plant Science*, 60(2), 695–701.
- [25] American Oil Chemists' Society (AOCS). (2017). AOCS Official Method Ba 9-58: Urease activity. *Official Methods and Recommended Practices of the AOCS (7th Ed.)*, American Oil Chemists' Society, Urbana, United States.
- [26] Mariotti, F., Tomé, D., & Patureau Mirand, P. (2008). Converting nitrogen into protein — beyond 6.25 and Jones' factors. *Critical Reviews in Food Science and Nutrition*, 48(2), 177–184. doi:10.1080/10408390701279749.
- [27] Food and Agriculture Organization of the United Nations (FAO), World Health Organization (WHO), & United Nations University (UNU). (2007). *Protein and Amino Acid Requirements in Human Nutrition*. WHO Technical Report Series 935, World Health Organization, Geneva, Switzerland. ISBN: 978-92-4-120935-9.
- [28] Schmidt, H. O., & de Oliveira, V. R. (2023). Overview of the incorporation of legumes into new food options: An approach on versatility, nutritional, technological, and sensory quality. *Foods*, 12(13), 2586. doi:10.3390/foods12132586.
- [29] Jribi, S., Choksi, A., & Besbes, S. (2021). Germination improves the nutritional composition and bioactive properties of chickpea and lentil flours. *Foods*, 10(11), 2827. doi:10.3390/foods10112827.
- [30] Du, Y., Yan, J., Wei, H., Xie, H., Wu, Y., & Zhou, J. (2023). Drying kinetics of paddy drying with graphene far-infrared drying equipment at different IR temperatures, radiation distances, grain-flow, and dehumidifying velocities. *Case Studies in Thermal Engineering*, 43, 102780. doi:10.1016/j.csite.2023.102780.
- [31] Chatchavanthatri, N., Junyusen, T., Arjham, W., Treeamnuk, T., Junyusen, P., & Pakawanit, P. (2021). Effects of parboiling and infrared radiation drying on the quality of germinated brown rice. *Journal of Food Processing and Preservation*, 45(11), e15892. doi:10.1111/jfpp.15892.
- [32] Marchini, M., Marti, A., Folli, C., Prandi, B., Ganino, T., & Conte, P. (2021). Sprouting of sorghum (*Sorghum bicolor* [L.] Moench): Effect of drying treatment on protein and starch features. *Foods*, 10(2), 407. doi:10.3390/foods10020407.
- [33] Sarjerao, L. S. K., Kashyap, P., & Sharma, P. (2022). Effect of drying techniques on drying kinetics, antioxidant capacity, structural and thermal characteristics of germinated mung beans (*Vigna radiata*). *Journal of Food Process Engineering*, 45(7), e14155. doi:10.1111/jfpe.14155.
- [34] Ahmed, M., Akter, M. S., & Eun, J.-B. (2011). Optimisation of drying conditions for the extraction of β -carotene, phenolic and ascorbic acid content from yellow-fleshed sweet potato using response surface methodology. *International Journal of Food Science & Technology*, 46(7), 1356–1362. doi:10.1111/j.1365-2621.2011.02632.x..
- [35] Giannakourou, M. C., & Taoukis, P. S. (2021). Effect of alternative preservation steps and storage on vitamin C stability in fruit and vegetable products: Critical review and kinetic modelling approaches. *Foods*, 10, 2630. doi:10.3390/foods10112630.
- [36] Goharpour, K., Salehi, F., & Daraei Garmakhany, A. (2024). Effects of different drying techniques of ground sprouted chickpeas on quality, textural properties, and sensory attributes of fried falafel. *Food Science & Nutrition*, 12, 6328–6337. doi:10.1002/fsn3.4240.
- [37] Berdejo, D., Pagán, E., Merino, N., García-Gonzalo, D., & Pagán, R. (2021). Emerging mutant populations of *Listeria monocytogenes* EGD-e under selective pressure of *Thymbra capitata* essential oil question its use in food preservation. *Food Research International*, 145, 110403. doi:10.1016/j.foodres.2021.110403.
- [38] Poulsen, M. W., Hedegaard, R. V., Andersen, J. M., de Courten, B., Bügel, S., Nielsen, J., Skibsted, L. H., & Dragsted, L. O. (2013). Advanced glycation endproducts in food and their effects on health. *Food and Chemical Toxicology*, 60, 10–37. doi:10.1016/j.fct.2013.06.052.

- [39] Najib, T., Heydari, M. M., & Meda, V. (2022). Combination of germination and innovative microwave-assisted infrared drying of lentils: Effect of physicochemical properties of different varieties on water uptake, germination, and drying kinetics. *Applied Food Research*, 2, 100040. doi:10.1016/j.afres.2021.100040.
- [40] Su, C., Saleh, A. S. M., Zhang, B., Feng, D., Zhao, J., & Guo, Y. (2020). Effects of germination followed by hot air and infrared drying on properties of naked barley flour and starch. *International Journal of Biological Macromolecules*, 165, 2060–2070. doi:10.1016/j.ijbiomac.2020.10.055.
- [41] Lignicka, I., & Galoburda, R. (2022). The role of cereal and pulse constituents and processing technology in flake texture development: A review. *Rural Sustainability Research*, 48(343), 34–44. doi:10.2478/plua-2022-0014.
- [42] Siitonen, A., Nieminen, F., Kallio, V., Tuccillo, F., Kantanen, K., Ramos-Diaz, J. M., Jouppila, K., Piironen, V., Kariluoto, S., & Edelmann, M. (2024). B vitamins in legume ingredients and their retention in high moisture extrusion. *Foods*, 13(5), 637. doi:10.3390/foods13050637.
- [43] Infantes-Garcia, M. R., Verkempinck, S. H. E., Del Castillo-Santaella, T., Maldonado-Valderrama, J., Hendrickx, M. E., & Grauwet, T. (2022). In vitro gastric lipid digestion of emulsions with mixed emulsifiers: Correlation between lipolysis kinetics and interfacial characteristics. *Food Hydrocolloids*, 128, 107576. doi:10.1016/j.foodhyd.2022.107576..
- [44] Rizvi, Q. U. E. H., Kumar, K., Ahmed, N., Yadav, A. N., Chauhan, D., Thakur, P., Jan, S., & Sheikh, I. (2022). Influence of soaking and germination treatments on the nutritional, anti-nutritional, and bioactive composition of pigeon pea (*Cajanus cajan* L.). *Journal of Applied Biology & Biotechnology*, 10(3), 100317. doi:10.7324/JABB.2022.100317.
- [45] Martín-Cabrejas, M. A., Ariza, N., Esteban, R., Mollá, E., Waldron, K., & López-Andréu, F. J. (2003). Effect of germination on the carbohydrate composition of the dietary fiber of peas (*Pisum sativum* L.). *Journal of Agricultural and Food Chemistry*, 51(5), 1254–1259. doi:10.1021/jf0207631.
- [46] Galal, W. K., Abd El-Salam, R. S., & Marie, A. M. (2024). High nutritional value instant flakes produced from various cereal grains. *Food Systems*, 7(1), 84–90. doi:10.21323/2618-9771-2024-7-1-84-90.
- [47] Singh, D. P., Singh, B., & Singh, D. (2025). Plant-based protein crops and their improvement: Current status and future perspectives. *Crop Science*, 65, e21389. doi:10.1002/csc2.21389.